Torque Ripple Verification in PM Machines

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Abstract — **In this paper, the problem of the ripple torque experienced with permanent magnet synchronous motors (PMSM) is discussed and solved by adopting advanced simulation tools. The two pulsating torque components are identified: the cogging (detent) torque and the slot-harmonics caused ripple. Simulated results are verified by using finite element method (FEM), taking into account the nonlinearity of magnetic material and secondary effects. The results are than proved on the experimental setup. Simplified model for the torque disturbance is proposed, suitable for designing active and passive compensating measures.**

Keywords — **Permanent magnet (PM) machines, cogging and ripple torque, feed-forward compensation**

I. Introduction

Mature technology of electrical drives offers widely accepted solutions for the power converter topology, basic control algorithms and most of the drive functions. Therefore, no significant changes in the drive structure will take place in the years to come. The structure of a typical motion control system includes ‘external loop’, dedicated to the mechanical subsystem, and ‘internal loop’ handling the motor flux and torque control [1]. Required bandwidth of the torque, speed and position loops are 1kHz, 200Hz and 60Hz. These requirements are especially needed for the high performance servo drives used in production machines and industrial robots.

The use of PMSM as servo actuator has great potential based on its energy-saving capabilities, high values of acceleration and relatively simple basic control structure [2], [3], [4], [5]. The drawback is two torque ripple components that are motor geometry dependent. Both are zero mean value nonlinear functions with multiple arguments, such arguments being the states of both the electrical and mechanical subsystem.

The first one, usually referred as ripple torque, is caused by the geometrically non-sinusoidal variation of air gap flux density and stator magnetomotive force (MMF) with rotor angle in electric degrees. The sixth and twelfth order harmonic is predominant [1], [6], [7]. On the other hand, cogging torque is current independent and it is the state of the rotor angle in mechanical degrees. The one is caused by the magnetic attraction of stator slotting and permanent magnet mounted on the rotor. The number of periods depends on the number of stator slots and pole pairs [1], [8], and [9].

The ripple and cogging torque is identified using simplified model based on the measurement performed on a typical synchronous servo motor with surface mount magnets. Simulated results are verified by using 2-D finite element method. The results are than proved on the experimental setup.

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II. Torque Pulsation First Simulation

The quantitative analysis of the electromagnetic torque takes usual assumptions that the end effects and the iron saturation and losses are negligible [10]. The first assumption allows the use of the superposition of the effects and strictly radial air gap flux lines, while the second one enable linearism of magnetic circuit. In this way, electromagnetic torque may be analyzed on the basis of the *Bli* and *Blv* law on each stator winding. Therefore, electromagnetic force (EMF) e produced in the stator windings in the slot with index j due to rotational external magnetic field is given by:

$$e_j = B_m \cdot L_j \cdot v \tag{1}$$

In the equation (1) L stands for the length of all slot windings of the same phase, v stands for the rotational speed and B_m for air gap flux density in front of the PM. The EMF for each phase (only phase a is presented) and electromagnetic torque are given by:

$$e_a = \sum_i^Q B_m \cdot L_{aj} \cdot v \tag{2}$$

$$T_e = \left(i_a \cdot e_a + i_b \cdot e_b + i_c \cdot e_c\right)/ w \tag{3}$$

where Q is the number of slots, i is stator current and w is angular speed. Since the real distribution of the air gap flux density and stator MMF is taken into account, the electromagnetic torque given in the equation (3) has both its nominal value and slot-harmonics caused ripple. The ripple torque represents its pulsating component.

The slotted machine brings stator anisotropy that causes the variation of the air gap magnetic energy. This produces the circumferential component of the attractive force ($\mathbf{F}_{cir}$) that attempts to minimize the system energy, that is to align the stator teeth and the PM. Each component of the force produces elementary cogging torque that is shown in Fig 1. The one may notice that total alignment of the slots and PM cancels elementary cogging torques resulting zero cogging torque of the PM. Therefore, this position is stated as magnetically stable.

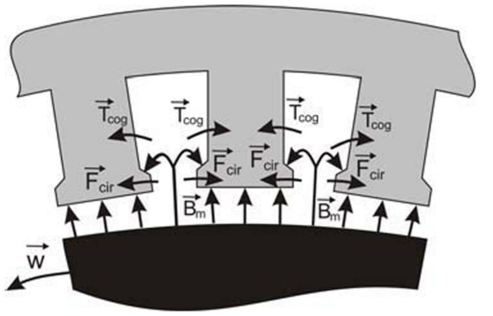

Fig 1: Elementary cogging torque generation: magnetically stable position

On the other hand, Fig 2 shows magnetically unstable position where the PM and stator slots are not aligned. The non-zero resulting cogging torque tends to move PM back into the stable position. Since its direction is opposite of the electromagnetic torque direction, the one is stated as negative.

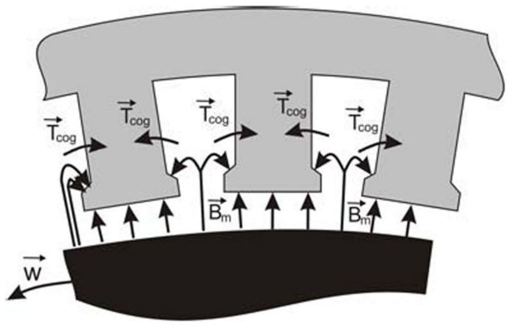

Fig 2: Magnetically unstable position

Cogging torque defines the reluctance torque given by

$$T_{cogg} = -\frac{1}{2} \cdot \Phi_g{}^2 \cdot \frac{d\Re_\Sigma}{d\theta} \tag{4}$$

where Φ_g is air gap flux and $\Re_\Sigma$ is overall air gap reluctance. The differential part of equation (4) may be presented as the reciprocal sum of the air gap permeances. Each addend is given as the algebraic sum of the slot and teeth air gap permeances ($\wp_1$ and $\wp_2$ respectively) in front of each PM. This is stated with following equations

$$\wp_k = \mu_0 \cdot (s_1 \cdot \wp_1 \cdot + s_2 \cdot \wp_2) = \mu_0 \cdot \left(s_1 \cdot L \cdot \frac{d_1}{l_{01}} + s_2 \cdot L \cdot \frac{d_2}{l_{02}} \right)$$

$$\Re_\Sigma = \frac{1}{\sum_k^{2p} \wp_k} \tag{5}$$

where $2p$ stands for the number of pairs (that is of the number of PMs), L stands for the machine length, while d is width, l_0 is length and parameter s is relative alignment with PM; index 1 denotes the teeth and 2 the slot. In the example given in Fig 2 s_1=2,8 and s_2=2,6.

The number of cogging torque waveform periods (N_{per}) depends on the number of cogging torque waveform periods per slot (N_p)

$$N_{per} = Q \cdot N_p, \quad N_p = \frac{2p}{\mathrm{HCF}(Q, 2p)} \tag{6}$$

where HCF stands for highest common factor [9].

In the case of different distances between PMs mounted on the rotor elementary torques are not evenly distributed along the air gap, meaning that not all magnets are in magnetically stable or unstable position at the same time. Since both positive and negative cogging torque affects the rotor, the resulting pulsation is decreased. The similar principle may be used in the case of the shaft made of rotor rings. The situation of the motor shaft realized of the rotor rings with different distances between PMs is presented in Fig 3.

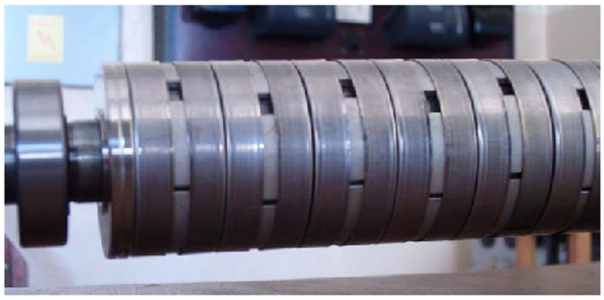

Fig 3: The shaft made of rotor rings

The analysis given by the equations (1)-(5) is used as the basis for the simulation written in language C. The results of the simulation are presented in section IV.

III. FEM SIMULATION

The simplifications in the quantitative analysis presented in section II are strictly radial air gap flux lines, linear magnetic circuit and rectangular stator teeth. Non ideally radial air gap flux lines are due to end effect, iron saturation and losses, and armature reaction. These affects air gap flux density both qualitative and quantitative: the magnitude of the flux density is not constant in front of the PM, and flux lines are not strictly radial. Further more, the effect of stator magnetic field is to increase the air gap flux density near the leading edge of each permanent magnet and decrease it near the lagging edge [1]. This implies that the results of the analysis given in section II should be confirmed by the finite element method simulation.

For that purpose the freeware Ansoft software Maxwell SV 9 is used [11], whose magnetostatic field solver enables computation of static magnetic field arising from DC currents and external magnetic sources such as PM. As results the one may analyze any quantity that can be derived from the basic magnetic field, including flux lines, forces and torques.

The effective data exchange between the C and FEM simulation was established in order to verify the results. Initial magnitude of stator current and appropriate rotor position necessary for the FEM simulation definition was given by the C simulation on the basis of approximated air gap flux density. The results in the form of the torque magnitude and air gap flux density are compared. Both simulations are repeated until this results closely matched.

IV. Simulation results

A. Non even PM distances

The main simulation parameters for the six pole, eighteen slot, surface mounted PMSM (rated torque T_n=7,3Nm; rated speed n_n=3000rpm) are B_m=0,85T and n=300rpm. The PM distances are not even and the motor shaft is made of one rotor ring only. The results of the cogging and ripple torques are presented in Fig 4 and 5. The magnitude and the shape of both torque pulsation components are as expected.

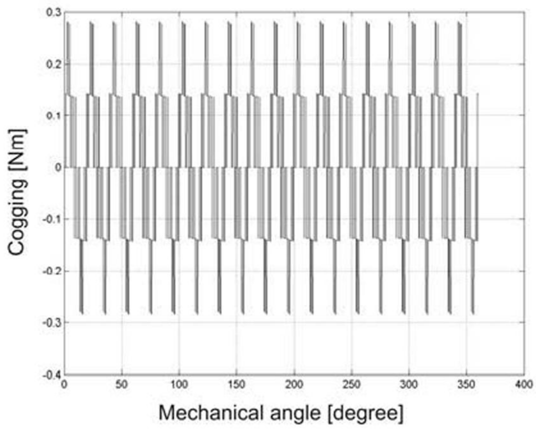

Fig 4: Cogging torque

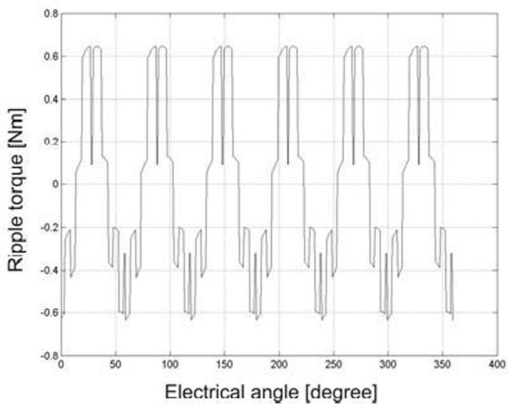

Fig 5: Ripple torque

B. The shaft made of rotor rings

The situation shown in Fig 3 is simulated, and the results are presented on Fig 6 9.

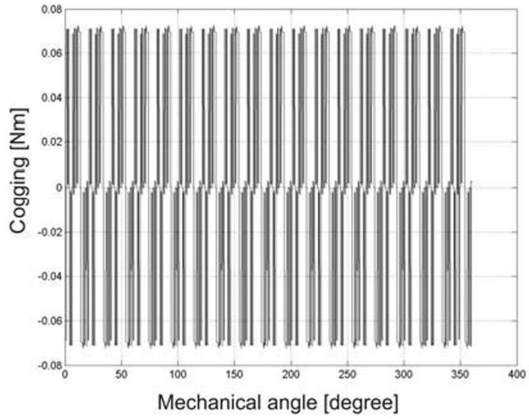

Fig 6: Cogging torque

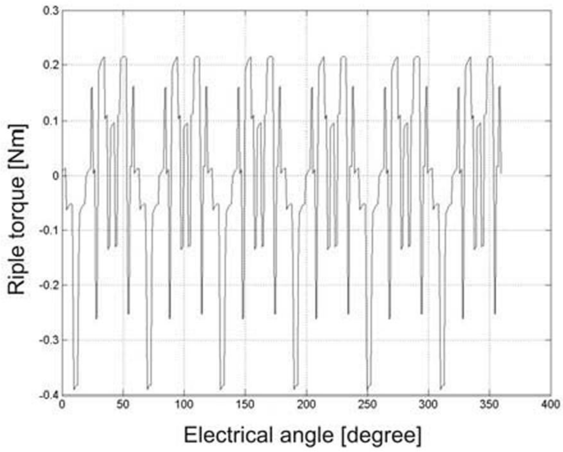

Fig 7: Ripple torque

The one may notice that the magnitude of both cogging and ripple torque presented in Fig 6 and 7 is considerably decreased. The pulsation torque due to motor geometry is the sum of cogging and ripple torque. The one is presented in Fig 8, while its harmonic content is presented in Fig 9. The magnitude of harmonic content is referenced by the magnitude of the highest harmonic component.

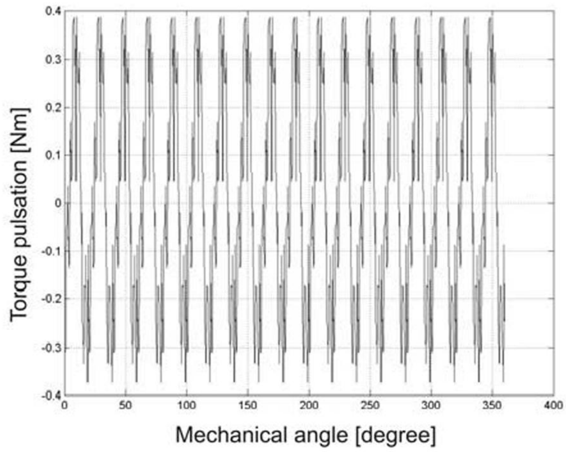

Fig 8: Pulsation torque

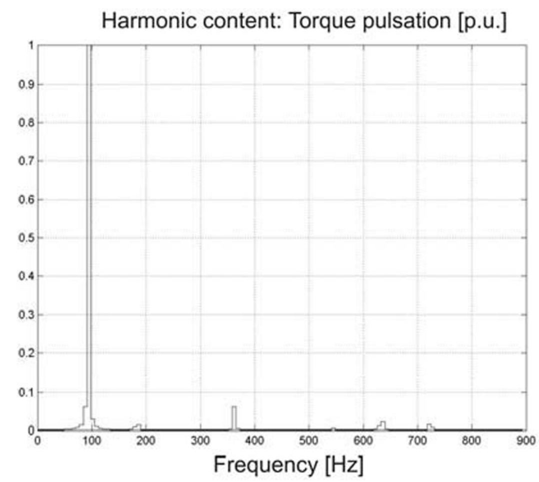

Fig 9: Pulsation torque harmonic content

C. FEM verification

The FEM simulation tested the torque production under various loads and conditions and the effects discussed in section III are proven. The influence of the stator reaction on the air gap flux density in the case of full load is presented in Fig 10. Even in this condition, the average value of air gap flux density the torque magnitude matched, which verified the results of the C simulation.

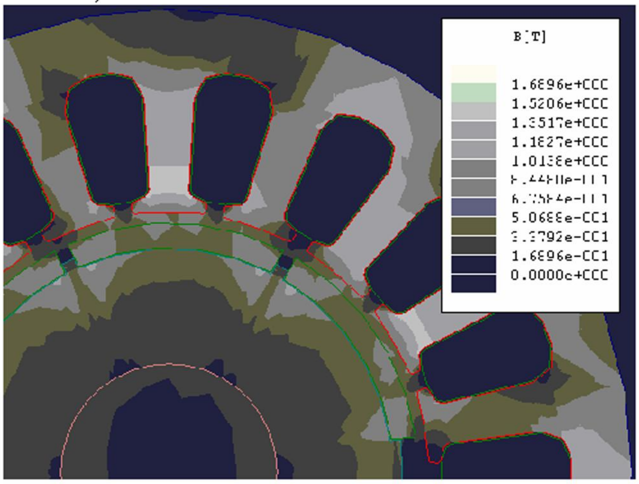

Fig 10: Flux density as the result air gap flux density matches the value used in C simulation

V. Experimental verification

The testing equipment consists of PMSM, electromagnetic resolver attached to the shaft and DBM03 servo amplifier [10]. The motor is speed controlled. The digital oscilloscope is used to measure and memorize *q*-component of the stator current which is directly proportional to the torque, while the resolver signals are decoded into the rotor position. The motor data are the same as in the C simulation. Measured signal encountered many sources of harmonics that are not connected to the motor geometry, such as the measuring method, quantization, the use of position feedback sensor [3] etc. Therefore, only expected harmonics are kept and the signal and its harmonic content are presented in Fig 11.

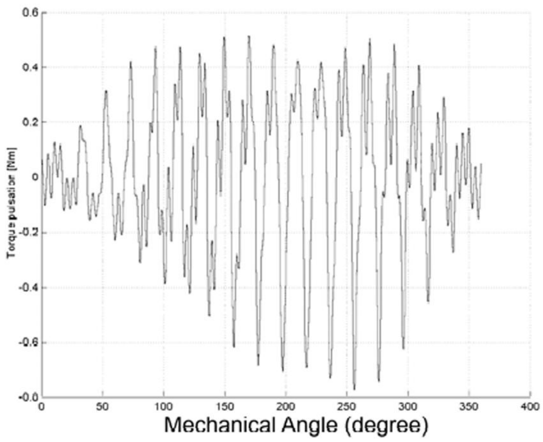

Fig 11: Testing results of the pulsation torque comparable to Fig 8

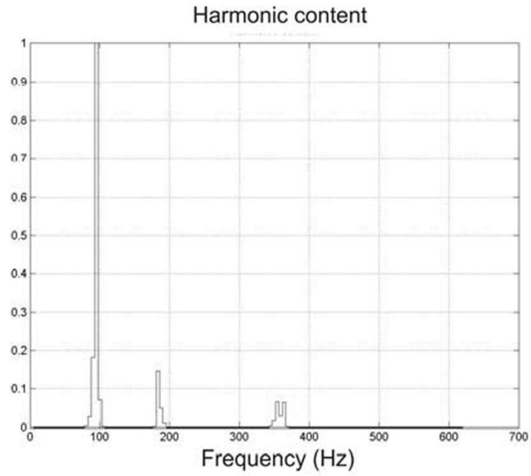

Fig 12: Pulsation torque harmonic content

VI. Conclusion

The contribution of this paper is the simplified model that identifies the torque ripple experienced with PMSM based on the measurement performed on a typical synchronous servo motor with surface mount magnets. Two pulsating components, the cogging torque and slot-harmonics caused ripple are discussed and solved by adopting advanced simulation tools. All geometric, electric and magnetic parameters of the machine are easily changeable, and the result is suitable for designing active and passive torque ripple compensation.

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